

Campus Biodiversity Survey
Environmental Science and Engineering
(EVS) Assignment -2

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1. Introduction

Biodiversity refers to the variety of life forms — plants, animals, fungi and microorganisms — present within a given area, along with the ecological complexes they form. A campus environment, despite being human-modified, often supports a surprisingly rich variety of flora and fauna, including flowering plants, insects, reptiles, birds and other invertebrates. This report presents the findings of a biodiversity survey conducted within the campus premises, with the objective of documenting the diversity of species present, understanding the ecological relationships among them, and appreciating the importance of even small green spaces in sustaining local ecosystems.

The survey was carried out through direct field observation and photography, recording species encountered in gardens, hedges, pavements, drains and tree canopies around the campus.

2. Study Area

Campus Name: SHIV NADAR UNIVERSITY CHENNAI

Date(s) of Survey: 10-08-2026 to 17-08-2026

Time of Survey: 8 days

The area surveyed included ornamental garden beds, flowering hedges, pathways, a small drain/water body, and tree cover within the campus, providing a mix of habitats such as leaf litter, foliage, open ground and standing water that support diverse plant and animal life.

3. Biodiversity Inventory

A total of 14 distinct species were recorded during the survey, spanning flowering plants, insects, reptiles, a bird and other invertebrates. The inventory below summarizes each species along with its category, ecological role and the specific location within the campus where it was observed.

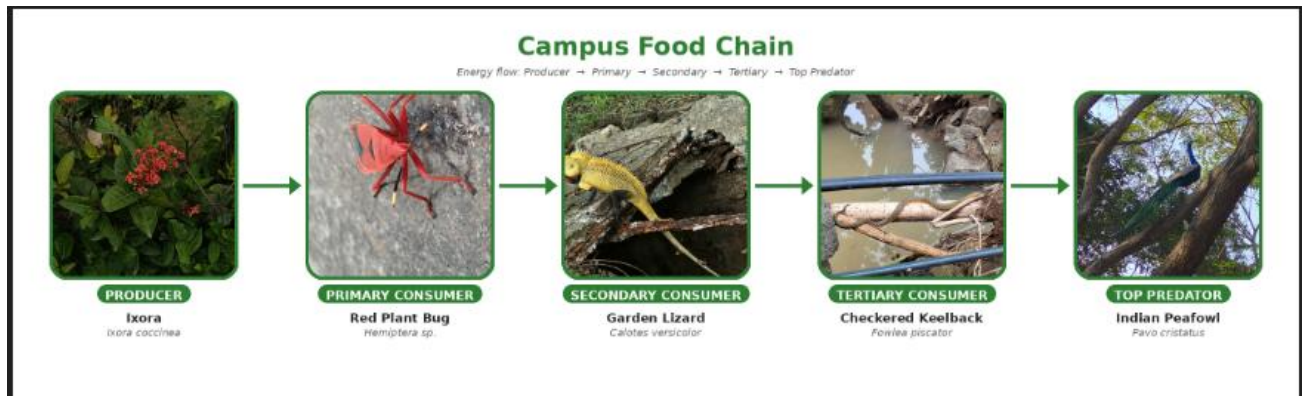
S.No	Common Name	Scientific Name	Category	Ecological Role
1	Crimson Rose Butterfly	Pachliopta hector	Insect (Lepidoptera)	Primary Consumer (nectar feeder / pollinator)
2	Jungle Flame (Ixora) - Red	Ixora coccinea	Plant (Angiosperm shrub)	Producer
3	Jungle Flame (Ixora) - Pink	Ixora coccinea (pink cultivar)	Plant (Angiosperm shrub)	Producer
4	Yellow Allamanda	Allamanda cathartica	Plant (Angiosperm climber/shrub)	Producer
5	Frangipani (Plumeria)	Plumeria rubra	Plant (Angiosperm tree/shrub)	Producer
6	Red Plant Bug (nymph)	Hemiptera sp.	Insect (Hemiptera)	Primary Consumer (plant-sap feeder)
7	Oriental Garden Lizard (breeding male)	Calotes versicolor	Reptile (Squamata)	Secondary Consumer (insectivore)

S.No	Common Name	Scientific Name	Category	Ecological Role
8	Oriental Garden Lizard (non-breeding)	Calotes versicolor	Reptile (Squamata)	Secondary Consumer (insectivore)
9	Indian Peafowl	Pavo cristatus	Bird (Galliformes)	Top Predator / Omnivore
10	Checkered Keelback (water snake)	Fowlea piscator	Reptile (Squamata)	Tertiary Consumer (carnivore)
11	Wandering Glider Dragonfly	Pantala flavescens	Insect (Odonata)	Secondary Consumer (predator of small insects)
12	Bougainvillea	Bougainvillea glabra	Plant (Angiosperm climber/shrub)	Producer
13	Rusty Millipede	Trigoniulus corallinus	Arthropod (Diplopoda)	Decomposer / Detritivore
14	Garden Snail	Land snail (Gastropoda sp.)	Mollusk (Gastropoda)	Primary Consumer / Detritivore

4. Food Chain

Using the organisms recorded during the survey, the following food chain can be constructed, representing a realistic feeding (trophic) relationship observed within the campus ecosystem:

Ixora (Producer) → Red Plant Bug (Primary Consumer) → Garden Lizard (Secondary Consumer) → Checkered Keelback Snake (Tertiary Consumer) → Indian Peafowl (Top Predator)



Trophic Level	Organism	Role
Producer	Ixora (<i>Ixora coccinea</i>)	Manufactures food via photosynthesis; base of the food chain.
Primary Consumer	Red Plant Bug (<i>Hemiptera sp.</i>)	Herbivorous insect that feeds directly on the Ixora plant's sap/tissue.
Secondary Consumer	Oriental Garden Lizard (<i>Calotes versicolor</i>)	Insectivorous reptile that preys on plant bugs and other small insects.
Tertiary Consumer	Checkered Keelback Snake (<i>Fowleia piscator</i>)	Carnivorous reptile that preys on garden lizards and other small vertebrates.
Top Predator	Indian Peafowl (<i>Pavo cristatus</i>)	Omnivorous bird known to prey on small snakes, completing the chain.

Ecological explanation: The Ixora shrub, as a photosynthetic producer, forms the energy base of this food chain. The Red Plant Bug, a primary consumer, feeds directly on the plant's tissues and sap, converting plant energy into animal biomass. The Oriental Garden Lizard, a secondary consumer, preys upon such insects, transferring energy up the chain. The Checkered Keelback Snake, a tertiary consumer, in turn preys on garden lizards (and other small vertebrates such as frogs) near the campus water body. Finally, the Indian Peafowl, acting as a top predator/omnivore in this chain, is known to prey opportunistically on small snakes, completing the energy transfer from producer to top predator. This chain illustrates how energy captured by plants during photosynthesis is progressively transferred through successive trophic levels, and highlights the interdependence of plant and animal life within the campus ecosystem.

5. Ecological Observations

The survey revealed a diverse assemblage of organisms occupying distinct micro-habitats within the campus:

- Flowering shrubs such as *Ixora*, *Allamanda*, *Plumeria* and *Bougainvillea* provide nectar and shelter, attracting pollinators such as the Crimson Rose butterfly and the Wandering Glider dragonfly.
- Ground-dwelling detritivores such as the Rusty Millipede and the Garden Snail were found among leaf litter and damp pavement, indicating their role in breaking down organic matter and recycling nutrients.
- Reptiles such as the Oriental Garden Lizard were observed basking on concrete kerbs and rocks, using these surfaces for thermoregulation while remaining close to vegetation that harbours their insect prey.
- The presence of a Checkered Keelback Snake near a shallow drain/water body indicates the availability of amphibian or small aquatic prey in that micro-habitat.
- The sighting of an Indian Peafowl in the tree canopy points to the presence of larger, more mobile fauna using the campus as part of a wider foraging range.

Overall, the observations suggest that even a relatively small, human-modified campus landscape can support multiple trophic levels — from primary producers to top predators — when adequate vegetation cover, leaf litter and water sources are available.

6. Photographic Documentation

The following section presents original photographs of each species recorded during the survey, along with a brief identification table for each.

Species 1: Crimson Rose Butterfly

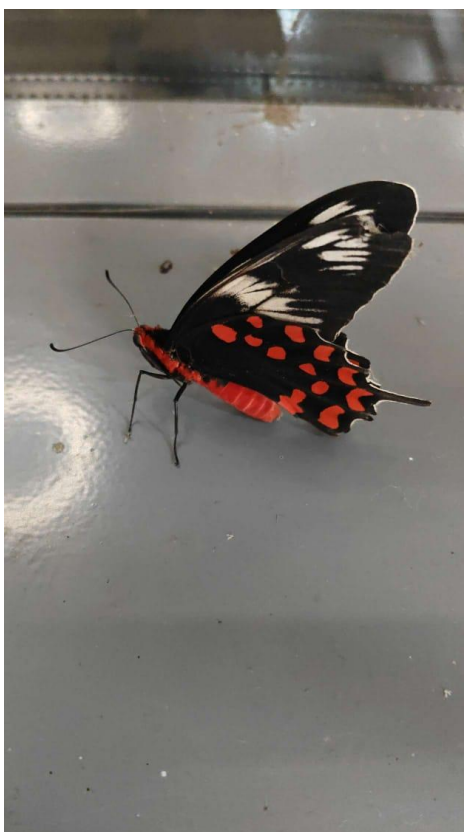


Fig. 1: Crimson Rose Butterfly (Pachliopta hector)

A swallowtail butterfly with black wings, white forewing band and crimson-red spotted hindwing and body; a common garden pollinator.

Common Name	Crimson Rose Butterfly
Scientific Name	Pachliopta hector
Category	Insect (Lepidoptera)
Ecological Role	Primary Consumer (nectar feeder / pollinator)
Location Observed	Gents hostel - 6

Species 2: Jungle Flame (Ixora) - Red



Fig. 2: Jungle Flame (Ixora) - Red (*Ixora coccinea*)

A flowering shrub bearing clusters of small tubular red-orange flowers; commonly used in ornamental hedges and attracts butterflies.

Common Name	Jungle Flame (Ixora) - Red
Scientific Name	<i>Ixora coccinea</i>
Category	Plant (Angiosperm shrub)
Ecological Role	Producer
Location Observed	AB-1 Entrance

Species 3: Jungle Flame (Ixora) - Pink



Fig. 3: Jungle Flame (Ixora) - Pink (*Ixora coccinea* (pink cultivar))

Pink-flowered cultivar of Ixora forming dense flowering hedges; an important nectar source for pollinating insects.

Common Name	Jungle Flame (Ixora) - Pink
Scientific Name	Ixora coccinea (pink cultivar)
Category	Plant (Angiosperm shrub)
Ecological Role	Producer
Location Observed	AB-1 Entrance

Species 4: Yellow Allamanda



Fig. 4: Yellow Allamanda (*Allamanda cathartica*)

A woody vine/shrub with whorled lance-shaped leaves and large funnel-shaped bright yellow flowers.

Common Name	Yellow Allamanda
Scientific Name	Allamanda cathartica
Category	Plant (Angiosperm climber/shrub)
Ecological Role	Producer
Location Observed	AB-1 Entrance

Species 5: Frangipani (Plumeria)



Fig. 5: Frangipani (Plumeria) (*Plumeria rubra*)

A small flowering tree with thick fragrant pink-and-white flowers and broad glossy leaves, widely planted for ornamentation and shade.

Common Name	Frangipani (Plumeria)
Scientific Name	<i>Plumeria rubra</i>
Category	Plant (Angiosperm tree/shrub)
Ecological Role	Producer
Location Observed	AB-1 Entrance

Species 6: Red Plant Bug (nymph)



Fig. 6: Red Plant Bug (nymph) (Hemiptera sp.)

A brightly coloured red bug nymph with long legs and banded antennae, typically found feeding on plant sap and seeds.

Common Name	Red Plant Bug (nymph)
Scientific Name	Hemiptera sp.
Category	Insect (Hemiptera)
Ecological Role	Primary Consumer (plant-sap feeder)
Location Observed	VAMA SUNDARI PARK

Species 7: Oriental Garden Lizard (breeding male)



Fig. 7: Oriental Garden Lizard (breeding male) (Calotes versicolor)

A male garden lizard in breeding colouration, showing a bright yellow body with a dark red-black throat patch; feeds mainly on insects.

Common Name	Oriental Garden Lizard (breeding male)
Scientific Name	Calotes versicolor
Category	Reptile (Squamata)
Ecological Role	Secondary Consumer (insectivore)
Location Observed	Well near mini Auditorium

Species 8: Oriental Garden Lizard (non-breeding)



Fig. 8: Oriental Garden Lizard (non-breeding) (Calotes versicolor)

The same species as above in its normal cryptic brown colouration, basking on a concrete kerb near leaf litter.

Common Name	Oriental Garden Lizard (non-breeding)
Scientific Name	Calotes versicolor
Category	Reptile (Squamata)
Ecological Role	Secondary Consumer (insectivore)
Location Observed	Near clock tower

Species 9: Indian Peafowl

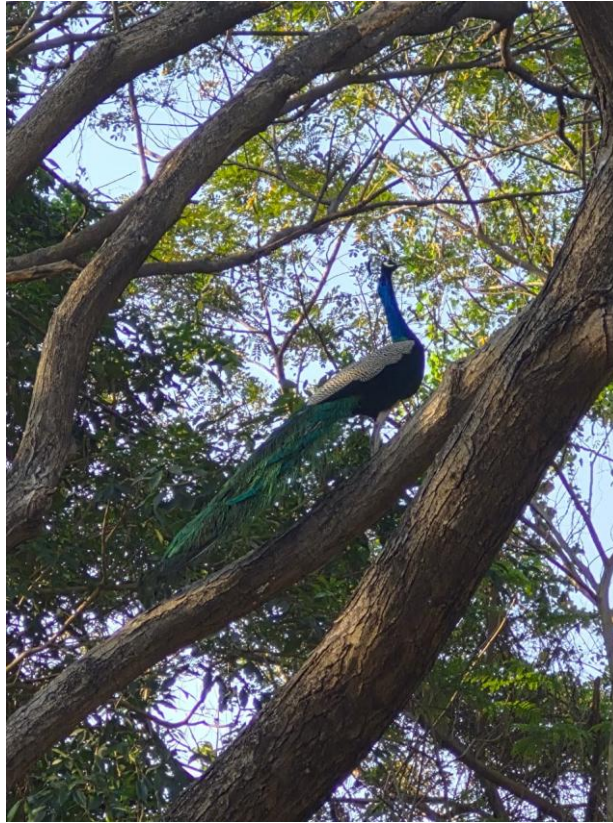


Fig. 9: Indian Peafowl (Pavo cristatus)

India's national bird, seen perched on a tree branch; an omnivore that feeds on insects, small reptiles, seeds and grains.

Common Name	Indian Peafowl
Scientific Name	Pavo cristatus
Category	Bird (Galliformes)
Ecological Role	Top Predator / Omnivore
Location Observed	Gents hostel main road

Species 10: Checkered Keelback (water snake)



Fig. 10: Checkered Keelback (water snake) (*Fowlea piscator*)

A non-venomous, yellow-headed, checkered-patterned water snake found near a shallow drain/pond, feeding on frogs and small vertebrates.

Common Name	Checkered Keelback (water snake)
Scientific Name	<i>Fowlea piscator</i>
Category	Reptile (Squamata)
Ecological Role	Tertiary Consumer (carnivore)
Location Observed	Gents hostel

Species 11: Wandering Glider Dragonfly

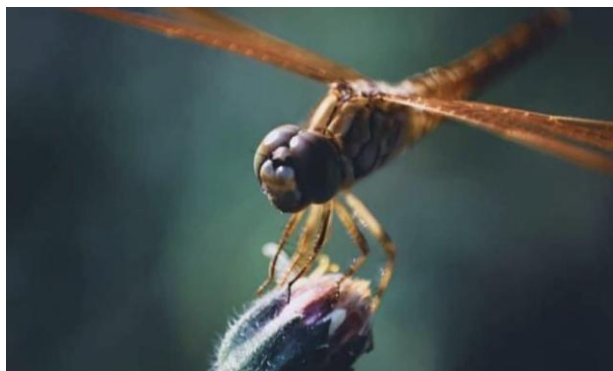


Fig. 11: Wandering Glider Dragonfly (*Pantala flavescens*)

A dragonfly captured in flight approaching a flower bud; an aerial predator that feeds on mosquitoes and other small insects.

Common Name	Wandering Glider Dragonfly
Scientific Name	Pantala flavescens
Category	Insect (Odonata)
Ecological Role	Secondary Consumer (predator of small insects)
Location Observed	CSE block _____

Species 12: Bougainvillea

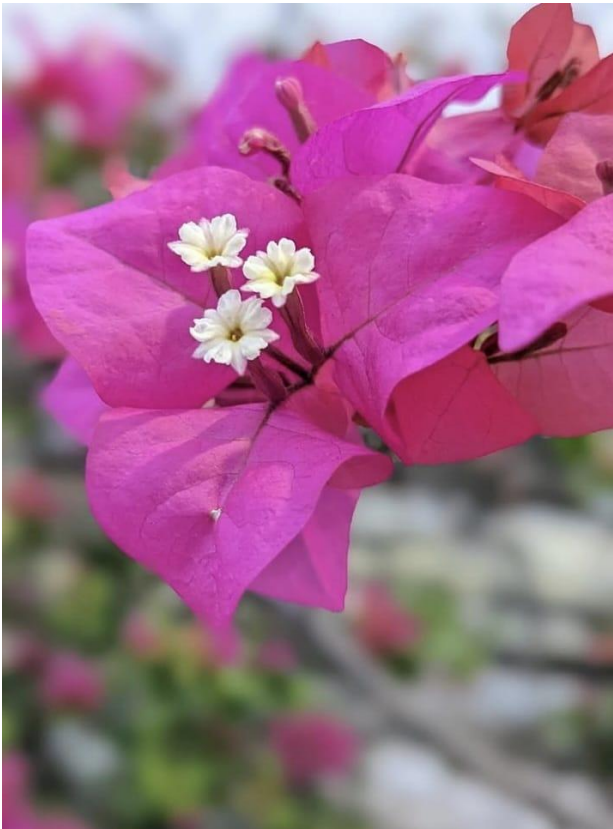


Fig. 12: Bougainvillea (*Bougainvillea glabra*)

A thorny ornamental vine with bright magenta-pink papery bracts surrounding small true white flowers.

Common Name	Bougainvillea
Scientific Name	Bougainvillea glabra
Category	Plant (Angiosperm climber/shrub)
Ecological Role	Producer
Location Observed	AB-3 Entrance

Species 13: Rusty Millipede



Fig. 13: Rusty Millipede (Trigoniulus corallinus)

A reddish-black banded millipede found on the ground; feeds on decaying leaf litter and plays a key role in nutrient recycling.

Common Name	Rusty Millipede
Scientific Name	Trigoniulus corallinus
Category	Arthropod (Diplopoda)
Ecological Role	Decomposer / Detritivore
Location Observed	Rishabhs

Species 14: Garden Snail



Fig. 14: Garden Snail (Land snail (Gastropoda sp.))

A terrestrial snail with a coiled shell, seen on wet pavement after rain; feeds on leaves, algae and decaying plant matter.

Common Name	Garden Snail
Scientific Name	Land snail (Gastropoda sp.)

Category	Mollusk (Gastropoda)
Ecological Role	Primary Consumer / Detritivore
Location Observed	OAT backside

7. Conclusion

This biodiversity survey of the campus documented 14 distinct species across plants, insects, reptiles, birds and other invertebrates, reflecting a healthy and interconnected micro-ecosystem within a limited urban/institutional space. The constructed food chain — from the *Ixora* plant through insect, reptilian and avian consumers — demonstrates how energy flows through successive trophic levels even within a small, managed landscape. Such surveys underline the ecological value of maintaining green cover, flowering plants and water bodies within campuses, as these features support pollinators, decomposers, predators and, ultimately, a self-sustaining food web. Conserving and expanding such green spaces can further enhance local biodiversity and contribute to a healthier campus environment.